

Overview

The Clinical Cholangiocarcinoma Biomarker Discovery Platform (CCA-BDP) is an end-to-end translational biomarker discovery system: from TCGA/GEO data acquisition through machine learning, external validation, explainable AI, biological interpretation, prognostic analysis, and therapeutic target discovery.

The deployed diagnostic model is a Random Forest classifier (500 trees, class_weight="balanced"), trained on 44 TCGA-CHOL samples using a 20-gene clinical biomarker panel selected for reproducibility across independent GEO cohorts, distinct from the broader nested cross-validation feature-stability pool used for methodological transparency.

Diagnostic Performance (External Validation)

Pooled across 444 samples from 3 independent GEO cohorts (GSE26566, GSE32225, GSE89749), the model achieves ROC-AUC=0.933, Sensitivity=0.111, Specificity=0.986.

GSE32225 and GSE89749 predict entirely class 0 (Normal) despite being mostly labeled Tumor; GSE26566 is the only cohort with a realistic mix of both predicted classes. This is reported plainly as a real limitation of cross-platform generalization, not smoothed over.

Cohort	Accuracy	ROC_AUC	Sensitivity	Specificity
GSE26566	0.6213	0.9355	0.3942	0.9846
GSE32225	0.0387	0.5	0.0	1.0
GSE89749	0.0167	0.5	0.0	1.0

Table 1. Per-cohort external validation metrics.

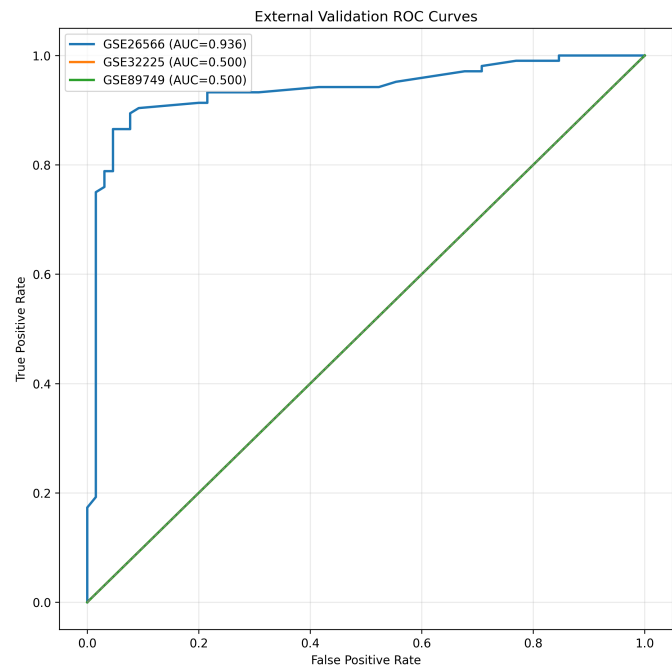


Figure 1. ROC curves per external cohort.

Explainable AI

SHAP (TreeExplainer, tree_path_dependent mode) identifies USH2A as the top globally important biomarker in the TCGA training population. Permutation importance computed on the held-out external cohorts diverges from training-based importance rankings (Spearman rho=0.20 vs. RF built-in

importance $\rho=0.98$), reflecting genuine distribution shift between training and external data rather than a computation error.

rank	gene_name	mean_abs_shap
1	USH2A	0.0391
2	ITIH1	0.0347
3	DGAT2	0.0346
4	JADE2	0.0318
5	OTC	0.0305
6	PON1	0.0295
7	RAD51	0.0292
8	MTHFD2	0.0287
9	MCF2L2	0.0253
10	SPDL1	0.0249

Table 2. Top 10 genes by global SHAP importance.

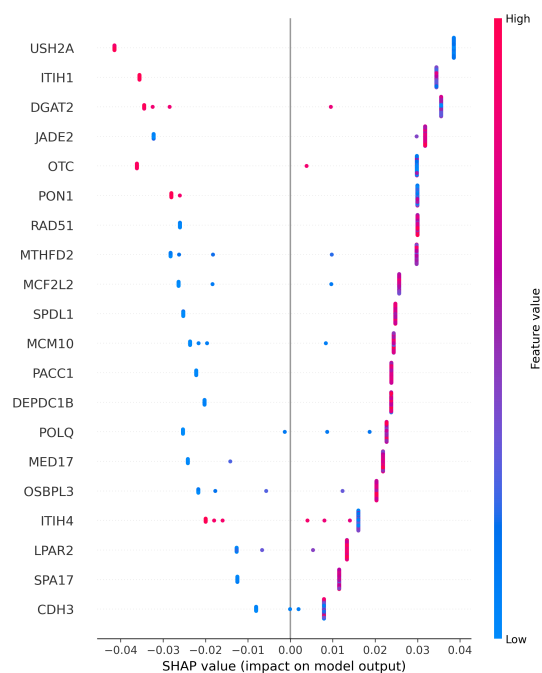


Figure 2. SHAP summary (training population).

Biological Validation

Gene Ontology Molecular Function enrichment reaches significance after multiple-testing correction (driven by JADE2 histone-acetyltransferase activity and RAD51/MCM10 DNA-binding terms); GO Biological Process, GO Cellular Component, KEGG, Reactome, and WikiPathways do not, consistent with a panel selected via independent statistical/ML consensus scoring rather than pathway curation.

The STRING protein-protein interaction network is sparse (7 edges among 20 genes, 5 hub genes), also expected for the same reason.

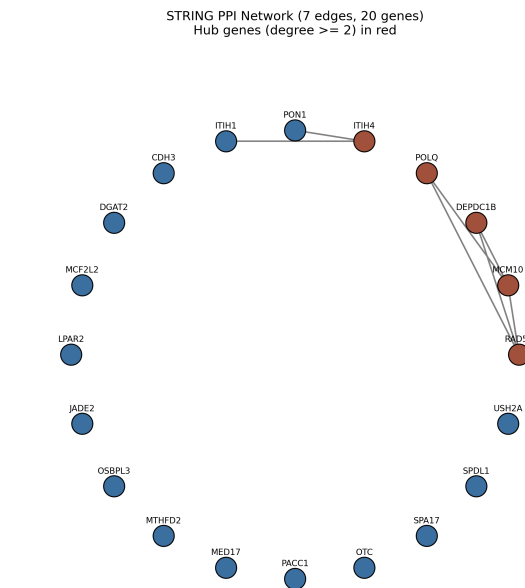


Figure 3. STRING protein-protein interaction network.

Survival & Prognostic Analysis

In an exploratory, hypothesis-generating analysis of 34 TCGA-CHOL tumor samples (18 observed deaths), CDH3 reached significance by Kaplan-Meier/log-rank test. Continuous univariate Cox regression did not reach conventional significance for any gene, a legitimate divergence at this sample size rather than a computational discrepancy. Given only 18 observed deaths, any multivariate model was deliberately restricted to at most 2 covariates.

gene_name	hazard_ratio	p_value
ITIH4	1.6817	0.0831
PACC1	1.5797	0.1212
LPAR2	0.7726	0.2814
MTHFD2	0.7674	0.2876
PON1	1.2651	0.3239

Table 3. Top 5 genes by univariate Cox hazard ratio.

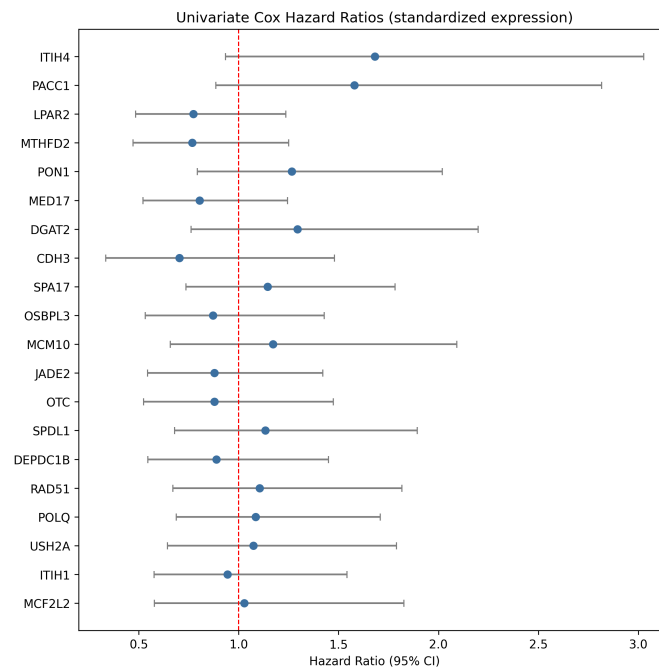


Figure 4. Univariate Cox hazard ratios.

Drug Target Discovery

A composite target prioritization score (equally weighting predictive importance, external-cohort SHAP relevance, biological relevance, and survival association) ranks DGAT2 highest. The top repurposing candidate is DGAT2 -> ERVOGASTAT (DGIdb), an existing compound with independent clinical development history now flagged as a candidate for this indication. DrugBank was not integrated (paid academic license required); DGIdb and Open Targets (both free) provide the drug-gene evidence.

gene_name	target_priority_score
DGAT2	0.6972
RAD51	0.6509
MTHFD2	0.5795
PON1	0.5782
USH2A	0.5685

Table 4. Top 5 prioritized drug targets.

Limitations

Key limitations, stated directly rather than minimized: (1) two of three external validation cohorts predict a single class, limiting the diversity of evidence for sensitivity in particular; (2) the survival cohort (34 patients, 18 events) is small and all survival findings are exploratory; (3) no enrichment analysis beyond GO Molecular Function reaches statistical significance after correction; (4) scikit-survival could not be installed in this environment, so time-dependent discrimination uses a simplified landmark-time method rather than full IPCW-weighted dynamic AUC.